An exact obstruction to the quadratic half-line completion

Abstract

The proposed order-33 cone is a proper subcone of the full nonnegative canonical valid-slack cone. We give an all-positive rational source and an explicit rational separating functional, certified by four positive definite dual matrices. A fixed enlargement, obtained by allowing a selector quartic in the square root of the Bob variable, contains the excluded source. Its order-49 affine pencil and exact feasible matrices are supplied. This proves strict enlargement, not global adequacy of the new family.

1 The two cones and the counterexample

Use the source order

$$c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \in \mathbb{R}^9_+, \qquad D_u = 1 + u^2, \quad D_v = 1 + v^2.$$

The full canonical cone $\mathcal{V}_9$ consists of the sources for which the established canonical numerator

$$\begin{aligned} F_c = {} & 4\lambda D_u (W + vX)^2 + 4nD_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v (uW - Y)^2 + 4AD_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d) D_u (vY - Z)^2 + (4\beta + 2d) D_v (X + uZ)^2 - 2dD_u D_v Z^2 \end{aligned} \tag{1}$$

is nonnegative for every real state vector and every pair of measurement angles, including their projective endpoints. Complex states are included by separating real and imaginary parts of this real symmetric quadratic form.

For clarity, the accepted two-LMI definitions are recorded here:

$$P = \begin{pmatrix} 4mv^2 & -4mv \\ -4mv & 4m + (4\beta + 2d)D_v \end{pmatrix}, \qquad N = \begin{pmatrix} 4CD_v + 4D + 2d & (4D + 2d)v \\ (4D + 2d)v & (4D + 2d)v^2 \end{pmatrix},$$

$$E = \begin{pmatrix} 4AD_v + (4\alpha + 2d)v^2 & -(4\alpha + 2d)v \\ -(4\alpha + 2d)v & 4\alpha - 2dv^2 \end{pmatrix}, \qquad K = 4\lambda \begin{pmatrix} 1 & v \\ v & v^2 \end{pmatrix} + 4nD_v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \tag{2}$$

$$G_J = \begin{pmatrix} P + K & J \\ J^T & N + K \end{pmatrix}, \qquad H_J = \begin{pmatrix} P + E & K - E - J \\ K - E - J^T & N + E \end{pmatrix}. \tag{3}$$

Let $\mathcal{C}_{33}$ be the cone obtained by requiring a quadratic selector $J(v) = J_0 + vJ_1 + v^2 J_2$ on $v \geq 0$. Equivalently, writing $Q(v) = Q_0 + vQ_1 + v^2 Q_2$ for $Q = G_J, H_J$, its four PSD blocks are

$$S_Q = \begin{pmatrix} Q_0 & B_Q \\ B_Q^T & Q_2 \end{pmatrix} \succeq 0, \qquad T_Q = Q_1 - B_Q - B_Q^T \succeq 0. \tag{4}$$

The nine source-sign blocks give total order $8 + 8 + 4 + 4 + 9 = 33$ and 44 auxiliaries. The accepted two-LMI identity gives $\mathcal{C}_{33} \subseteq \mathcal{V}_9$. Reflection by $T = \mathrm{diag}(1, -1)$ supplies the negative half-line; homogeneous continuity supplies the projective endpoints. No restriction on state signs is made.

Theorem 1. *The all-positive rational source*

$$\boxed{\bar{c} = \left(1, 2^{-34}, 2^{-14}, 2^{-24}, \frac{1}{576}, 161, \frac{1}{2}, 76, \frac{1}{250}\right)} \tag{5}$$

belongs to $\mathcal{V}_9$ but not to $\mathcal{C}_{33}$. Moreover, the fixed quartic-in-$\sqrt{v}$ completion cone $\mathcal{C}_{\mathrm{ram}}$ defined in Section 4 satisfies

$$\boxed{\mathcal{C}_{33} \subsetneq \mathcal{C}_{\mathrm{ram}} \subseteq \mathcal{V}_9, \qquad \bar{c} \in \mathcal{C}_{\mathrm{ram}} \setminus \mathcal{C}_{33}.}$$

It has an evaluated rational affine pencil of order 49 with nine source coordinates and 148 auxiliaries. Equality $\mathcal{C}_{\mathrm{ram}} = \mathcal{V}_9$ is not asserted.

2 The exact separating inequality

Here is the integer linear functional that separates (5):

$$\begin{aligned}\ell(c) = \; & -23803618d + 906393494300170\lambda + 2744379904n \\ & + 1378590360912m + 5935589516C + 20425A \\ & + 2600028D + 91248\alpha + 278104928\beta.\end{aligned} \tag{6}$$

Its exact value is

$$\boxed{\ell(\bar{c}) = -\frac{5417798011632274183}{9663676416000} < -560635.} \tag{7}$$

We prove $\ell \geq 0$ on the entire cone $\mathcal{C}_{33}$, rather than only on a collection of test instances.

Lemma 2 (Dual trace identity)**.** *For $\sigma = G, H$, let $U_\sigma, R_\sigma, W_\sigma$ be symmetric 4×4 matrices satisfying*

$$M_\sigma = \begin{pmatrix} U_\sigma & R_\sigma \\ R_\sigma & W_\sigma \end{pmatrix} \succeq 0, \qquad R_\sigma \succeq 0. \tag{8}$$

For each of the three pairs (U_G, U_H), (R_G, R_H), (W_G, W_H), require their upper-right 2×2 blocks to agree. These are twelve scalar equations. Put $M_{\sigma,0} = U_\sigma$, $M_{\sigma,1} = R_\sigma$, $M_{\sigma,2} = W_\sigma$. Then

$$f(c) = \sum_{k=0}^{2} [\langle M_{G,k}, G_k(c,0)\rangle + \langle M_{H,k}, H_k(c,0)\rangle] \tag{9}$$

is nonnegative on $\mathcal{C}_{33}$. Here $G_k(c,0), H_k(c,0)$ mean the coefficients with $J = 0$, and $\langle A, B\rangle = \operatorname{tr}(AB)$.

Proof. For each side, expand (4) to obtain

$$\langle M_\sigma, S_\sigma\rangle + \langle R_\sigma, T_\sigma\rangle = \langle U_\sigma, Q_0\rangle + \langle R_\sigma, Q_1\rangle + \langle W_\sigma, Q_2\rangle.$$

The unrestricted B_σ cancels because R_σ is symmetric. The coefficients J_k enter G_k, H_k through opposite off-diagonal blocks, so the twelve overlap equations cancel every entry of J_0, J_1, J_2. Consequently

$$f(c) = \langle M_G, S_G\rangle + \langle R_G, T_G\rangle + \langle M_H, S_H\rangle + \langle R_H, T_H\rangle \geq 0. \tag{10}$$

This is a direct trace identity; no dual-attainment assertion is needed. □

Appendix A displays six rational 4×4 matrices satisfying (8) and the twelve overlap equations, with

$$f(c) = 2^{-42}\ell(c). \tag{11}$$

Their four PSD blocks are in fact positive definite. The source-sign dual variables are all zero. Equations (7)–(11) therefore prove $\bar{c} \notin \mathcal{C}_{33}$ by finite rational arithmetic.

This also excludes a polynomial half-line selector of any degree. Indeed, every 2×2 principal minor of G_J gives

$$|J_{ij}(v)|^2 \leq (P+K)_{ii}(v)(N+K)_{jj}(v) = O(v^4) \qquad (v \longrightarrow +\infty).$$

Every polynomial entry must therefore have degree at most two, already excluded by the certificate above. The obstruction requires genuinely nonpolynomial dependence on the native variable v.

3 Validity of the source

We reuse the established rational source

$$\begin{aligned}(d, \lambda, n) &= (1, 0, 0), \\ (m, C, A) &= \left(\frac{1166523000}{31661820415017653}, \frac{322803000}{186199784281}, \frac{218288946200}{1359122513}\right), \\ (D, \alpha, \beta) &= \left(\frac{90943434293}{222507771414}, \frac{248802430236}{3300726103}, \frac{205476973}{51646655494}\right),\end{aligned} \tag{12}$$

denoted c_*. Its exact contact is $(u,v,\psi)=(1/13,16,(140,-1/32,1/50,1))$. The public source-validity certificate is retained verbatim in the supplied data. For a self-contained analytical check, let $\Pi_*(u,v)$ be the 4×4 state matrix of (1), and set $x=169u^2$, $y=v^2/256$. The retained determinant identity can be written

$$\det\Pi_* = 16\kappa D_u^2 D_v^2 u^2 v^2\,\mathcal{A}(x,y),\quad \kappa=\frac{365997345792000000}{80653488310452717062149398114218073953704585845 7}>0, \tag{13}$$

where

$$\begin{aligned}\mathcal{A}(x,y) &= \rho_1(x+x^{-1}-2)+\rho_2(y+y^{-1}-2)\\ &\quad+\rho_3(y/x+x+y^{-1}-3)+\rho_4(xy+x^{-1}+y^{-1}-3),\end{aligned}$$

and the positive integer coefficients are

ρ_1	126791783324459354102518032323 1
ρ_2	762274793296209851400745596627083
ρ_3	273779118730780662039479999 6019
ρ_4	791658120509994203488525245 00

Each bracket is nonnegative by the two- or three-term AM–GM inequality. To pass from determinant positivity to all-state positivity, the $Z=0$ principal block B_* satisfies

$$B_*\succeq T_*^T\,\mathrm{diag}(4m,4CD_v,4AD_uD_v)T_*\succ 0,\qquad T_*=\begin{pmatrix}v&-1&uv\\u&0&-1\\0&0&1\end{pmatrix},\quad \det T_*=u \tag{14}$$

when $u\neq 0$. All omitted terms of (1) are nonnegative on $Z=0$. Thus the scalar Schur complement and (13) give $\Pi_*\succeq 0$ for $uv\neq 0$. Polynomial continuity recovers the axes; bihomogenization in the two angle pairs recovers every projective endpoint. This is the full state matrix argument, not a determinant-only test.

Direct rational comparison gives $\bar{c}-c_*\geq 0$, with the d coordinate unchanged. Every other coordinate in (1) multiplies a nonnegative event form. Therefore

$$F_{\bar{c}}=F_{c_*}+F_{\bar{c}-c_*}\geq 0$$

for all states and tests. This proves $\bar{c}\in\mathcal{V}_9$. The unrounded contact source itself is also separated:

$$\ell(c_*)=-\frac{30372270056956512321138766804 0}{267603838100324717114473}<0. \tag{15}$$

4 One fixed enlargement that repairs the obstruction

Set $v=16s^2$ on the positive half-line. Allow a selector $J(s)=\sum_{k=0}^4 J_k s^k$. This adds the selector powers $\sqrt{v}$ and $v^{3/2}$; their coefficients are auxiliary variables, not functions of the test. Put

$$Z_2(s)=\begin{pmatrix}I_4\\sI_4\\s^2I_4\end{pmatrix},\qquad Z_1(s)=\begin{pmatrix}I_4\\sI_4\end{pmatrix}.$$

For $Q=G_J,H_J$, impose the identity

$$Q(16s^2)=Z_2(s)^TA_QZ_2(s)+sZ_1(s)^TB_QZ_1(s),\qquad A_Q\succeq 0,\quad B_Q\succeq 0. \tag{16}$$

Here the notation on the left means substitution in the source blocks, with $J=J(s)$. The block orders are $12,8,12,8$ and nine scalar source blocks. This defines $\mathcal{C}_{\mathrm{ram}}$ and immediately proves $\mathcal{C}_{\mathrm{ram}}\subseteq\mathcal{V}_9$ using the accepted two-LMI identity. Matrix half-line factorization also shows that (16) captures exactly the quartic-in-s selector condition, including singular matrices. Substituting $v=16s^2$ in any quadratic half-line certificate gives one of these Grams, proving $\mathcal{C}_{33}\subseteq\mathcal{C}_{\mathrm{ram}}$.

An actual affine pencil. There is no unresolved coefficient-matching problem in this enlargement. For a prescribed quartic $Q(s) = \sum_{k=0}^{4} Q_k s^k$, write $A = (A_{ij})_{i,j=0}^{2}$ and $B = (B_{ij})_{i,j=0}^{1}$ in 4×4 blocks. Choose B_{00}, B_{11} symmetric, B_{01}, A_{02} arbitrary, and K_{01}, K_{12} skew-symmetric. Set

$$\begin{aligned} A_{00} &= Q_0, & A_{22} &= Q_4, \\ A_{01} &= \tfrac{1}{2}(Q_1 - B_{00}) + K_{01}, & A_{12} &= \tfrac{1}{2}(Q_3 - B_{11}) + K_{12}, \\ A_{11} &= Q_2 - A_{02} - A_{02}^T - B_{01} - B_{01}^T. \end{aligned} \tag{17}$$

Transpose the lower blocks. These formulas parametrize every coefficient solution of (16). They use $10 + 10 + 16 + 16 + 6 + 6 = 64$ variables per side, in addition to the 20 selector coefficients. Thus the final pencil has 148 auxiliaries and no coefficient equations. The exported 49-dimensional pencil has 411 nonzero upper-triangular rational coefficient entries, with every source and auxiliary coefficient evaluated.

Exact repair of both sources. The supplied data give rational face matrices V_i and positive definite rational matrices X_i of orders $7, 3, 11, 7$. Their congruences give the four Gram matrices in (16) at c_*. Every unpivoted LDL pivot of X_i is greater than 2^{-18}; all 80 coefficient conditions after selector elimination are checked exactly. The full rational Grams for $\bar{c}$ are also exported, together with their 148 affine-pencil auxiliaries. Their PSD proof is the same four positive congruences plus nonnegative rational event outer products.

The extension from c_* to $\bar{c}$ is elementary. For $\delta = \bar{c} - c_*$, one has $d_\delta = 0$, and $P_\delta, N_\delta, E_\delta, K_\delta$ are sums of polynomial event outer products. Taking $J_\delta = K_\delta$ gives

$$\begin{aligned} G_\delta &= \operatorname{diag}(P_\delta, N_\delta) + \begin{pmatrix} K_\delta & K_\delta \\ K_\delta & K_\delta \end{pmatrix}, \\ H_\delta &= \operatorname{diag}(P_\delta, N_\delta) + \begin{pmatrix} E_\delta & -E_\delta \\ -E_\delta & E_\delta \end{pmatrix}. \end{aligned}$$

These additions lie in the same quartic-in-s family. This also proves that the entire $d = 0$ source face is included, with arbitrary zero event weights. For example, the new selector at c_* has the nonzero odd coefficient

$$[s]J_{21}(s) = \frac{4449008858798426042034668223711 09}{3909264470979595000000000000000000}.$$

The explicit certificate uses odd powers of s; it is not another quadratic-in-v selector. The separation proves that at least one function outside that quadratic span is necessary. It does not prove uniqueness or minimality of the particular enlargement chosen here.

The fixed scalar testing family. For an explicit connection to the canonical slack, use the accepted identity with $\psi = (W, X, Y, Z)^T$,

$$\eta = L(u)\psi, \quad L(u) = \begin{pmatrix} u & 0 & u^2 & 0 \\ 0 & u & 0 & u^2 \\ u & 0 & -1 & 0 \\ 0 & u & 0 & -1 \end{pmatrix}, \quad U = u^2, \quad D_U = \operatorname{diag}(1, 1, U, U).$$

For $v \neq 0$ put $s = \sqrt{|v|/16}$ and $\eta_\sigma = \operatorname{diag}(1, \sigma, 1, \sigma)\eta$, where $\sigma = \operatorname{sgn}(v)$. At $v = 0$ choose $\sigma = 1$. The four groups of fixed functions are

$$\boxed{Z_2(s)D_U\eta_\sigma, \quad \sqrt{s}\, Z_1(s)D_U\eta_\sigma, \quad uZ_2(s)\eta_\sigma, \quad u\sqrt{s}\, Z_1(s)\eta_\sigma.} \tag{18}$$

They have sizes $12, 8, 12, 8$ and their block Gram form equals $u^2 D_u F_c$. The Bob factors are exactly $\{1, s, s^2, \sqrt{s}, s^{3/2}\}$, so the added Gram factors are fixed quarter powers of $|v|$. Source and testing variables remain independent. Dividing every function by $D_u^{3/2}\sqrt{D_v}$ represents

$$\frac{u^2}{D_u}\,\frac{F_c}{D_u D_v}. \tag{19}$$

This division involves only testing variables. Here is a literal compact algebraic cover, including every endpoint. Take

$$a_0^2 + a_1^2 = 1, \quad a_0 \geq 0, \qquad \rho^2 + \tau^2 = 1, \quad \rho, \tau \geq 0, \qquad \epsilon \in \{-1, 1\},$$

and put

$$\kappa = \sqrt{\tau^8 + 256\rho^8}, \quad b = \frac{(\tau^4, 16\epsilon\rho^4)}{\kappa}, \quad g = \frac{(\tau^4, \rho^2\tau^2, \rho^4)}{\kappa}, \quad r = \frac{(\rho\tau^3, \rho^3\tau)}{\kappa}.$$

These are source-independent algebraic functions with $\kappa^2 \geq \tau^8 + \rho^8 \geq 1/8$. In the interiors, $u = a_1/a_0$, $v = 16\epsilon(\rho/\tau)^4$. Define the homogeneous cubic vectors

$$P_G = \begin{pmatrix} a_0^2 a_1 W + a_0 a_1^2 Y \\ \epsilon(a_0^2 a_1 X + a_0 a_1^2 Z) \\ a_1^3 W - a_0 a_1^2 Y \\ \epsilon(a_1^3 X - a_0 a_1^2 Z) \end{pmatrix}, \qquad P_H = \begin{pmatrix} a_0 a_1^2 W + a_1^3 Y \\ \epsilon(a_0 a_1^2 X + a_1^3 Z) \\ a_0 a_1^2 W - a_0^2 a_1 Y \\ \epsilon(a_0 a_1^2 X - a_0^2 a_1 Z) \end{pmatrix}.$$

The forty functions on this closed cover are

$$g \otimes P_G, \quad r \otimes P_G, \quad g \otimes P_H, \quad r \otimes P_H.$$

Their Gram sum is $a_1^2 f_c(a, b; \psi)$, where f_c is the bihomogeneous normalized physical slack. This follows from (19) on the interior and is an algebraic identity everywhere. Each function has absolute value less than 3 when $\|\psi\| = 1$. At Bob's endpoints, $(g, r) = ((1, 0, 0), 0)$ or $((0, 0, 1/16), 0)$; the Alice endpoints are literal polynomial substitutions in P_G, P_H. The multiplier a_1^2 is positive on a dense testing set. Consequently every state sign and every endpoint is retained in the validity converse.

The precise unresolved implication. The question settled here is $\mathcal{V}_9 \subseteq \mathcal{C}_{33}$, which is false. The next unproved implication would be $c \in \mathcal{V}_9 \Rightarrow c \in \mathcal{C}_{\text{ram}}$: equivalently, every valid canonical source would need a selector in the one fixed span $\{1, s, s^2, s^3, s^4\}$. The two exact repaired sources do not establish that statement. The present order-49 pencil is an affine certificate for $\mathcal{C}_{\text{ram}}$, not a lift of the full canonical polar or of the full eight-coordinate quantum set. The existing complete companion is not replaced by this result.

A Six rational matrices proving the separation

Put $q = 2^{30}$. The following are the hatted moment matrices. The lower triangles are displayed as well so that there is no storage convention ambiguity.

$$q\widehat{U}_G = \begin{pmatrix} 3376576 & -4595 & 8890485/2^{20} & -141059/2^{14} \\ -4595 & 82702 & 374101/2^7 & -2966 \\ 8890485/2^{20} & 374101/2^7 & 48657 & -1435456 \\ -141059/2^{14} & -2966 & -1435456 & 159900862 \end{pmatrix},$$

$$q\widehat{R}_G = \begin{pmatrix} 2570 & 373 & 100461333/2^{23} & -796959/2^{16} \\ 373 & 5619 & 132105/2^5 & -4192 \\ 100461333/2^{23} & 132105/2^5 & 4654 & -50824 \\ -796959/2^{16} & -4192 & -50824 & 1442242 \end{pmatrix},$$

$$q\widehat{W}_G = \begin{pmatrix} 20 & 26 & 156439171/2^{23} & -18103/2^{10} \\ 26 & 822 & 92585/2^7 & -4442 \\ 156439171/2^{23} & 92585/2^7 & 697 & -4948 \\ -18103/2^{10} & -4442 & -4948 & 53164 \end{pmatrix},$$

$$q\widehat{U}_H = \begin{pmatrix} 142545776 & 374885 & 142247760 & -564236 \\ 374885 & 1002 & 374101 & -1483 \\ 142247760 & 374101 & 141950400 & -563057 \\ -564236 & -1483 & -563057 & 2251 \end{pmatrix},$$

$$q\widehat{R}_H = \begin{pmatrix} 201343592 & 529527 & 200922666 & -796959 \\ 529527 & 1401 & 528420 & -2096 \\ 200922666 & 528420 & 200502635 & -795292 \\ -796959 & -2096 & -795292 & 3163 \end{pmatrix},$$

$$q\widehat{W}_H = \begin{pmatrix} 313472891 & 94149 & 312878342 & -1158592 \\ 94149 & 16684 & 92585 & -2221 \\ 312878342 & 92585 & 312285067 & -1156239 \\ -1158592 & -2221 & -1156239 & 4509 \end{pmatrix}.$$

Set

$$\Delta_G = \mathrm{diag}(128, 1/32, 1/2, 1/32), \qquad \Delta_H = \mathrm{diag}(1/512, 1/16, 1/512, 1/32).$$

The moments used in Lemma 2 are

$$U_\sigma = \Delta_\sigma \widehat{U}_\sigma \Delta_\sigma, \quad R_\sigma = 16\Delta_\sigma \widehat{R}_\sigma \Delta_\sigma, \quad W_\sigma = 256\Delta_\sigma \widehat{W}_\sigma \Delta_\sigma.$$

The twelve upper-right-block equalities follow entry by entry. For positive definiteness, apply the rational recurrence

$$p_j = A_{jj} - \sum_{k<j} L_{jk}^2 p_k, \qquad L_{ij} = \frac{A_{ij} - \sum_{k<j} L_{ik} L_{jk} p_k}{p_j}, \quad L_{jj} = 1.$$

For $\widehat{M}_\sigma = \left(\begin{smallmatrix} \widehat{U}_\sigma & \widehat{R}_\sigma \\ \widehat{R}_\sigma & \widehat{W}_\sigma \end{smallmatrix}\right)$, the exact floors of $2^{30} p_j$ are

$\widehat{M}_G$	$(3376576, 82695, 48553, 117468710, 16, 14, 14, 31)$
$\widehat{R}_G$	$(2570, 5564, 1593, 10646)$
$\widehat{M}_H$	$(142545776, 16, 32, 17, 29078313, 12, 24, 11)$
$\widehat{R}_H$	$(201343592, 8, 15, 8)$.

The first scaled pivot in every row equals the listed integer; each later pivot lies strictly between its listed integer and that integer plus one. In particular, every pivot is greater than 2^{-27}. The supplied rational LDL factors give exact equalities, not numerical approximations. Finally, direct coefficient pairing with (2) gives (11). These six small matrices, the stated recurrence, and one rational substitution prove the separation.

Source and reproducibility

The source-validity identity is retained from the public project at commit

`b2bfe64cd81a951b7bbe33a84b9dedd5d4ba162c`,

under `extracted-lifts/canonical-completion/`. The new package contains the rational dual matrices and LDL factors, both repaired-source Gram certificates, the evaluated sparse affine pencil and feasible auxiliaries, the fixed function and product specifications, and standard-library exact verifiers. Run `python3 reproduce.py` from the package root. No SDP solver is required to verify any claimed result. The order-49 pencil here concerns the ramified two-LMI completion; it is distinct from the previously solved order-49 Laurent-coefficient cone.